

D858

# Secondary Earth Science Curriculum

Armored Drill

8 concepts · 12 items · 4 transfer items

**What this is.** Every concept in this course's public description, re-asked with more disguise than a practice test uses. Each stem carries facts that do not decide it. Every wrong answer is a named trap.

**Source.** Scope from the course's public description in the WGU Institutional Catalog (September 2026); items written from the field's standard body of knowledge. No WGU assessment item, course material or proprietary content is used or reproduced. Not affiliated with or endorsed by Western Governors University.

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## How to run it

Paper first. Pencil. One pass, no notes, a timer if you want the pressure.

1. **Before you read the options**, fill the strip under the item: the target word (what is actually being asked), and the one fact in the stem that decides it.
2. **Principle before arithmetic**. Name the rule, then compute.
3. **Camouflage**. Most of every stem does not matter. Ask: which single fact decides this?
4. **Inverse pairs are the bait**. If two options are opposites, one of them is the trap built for you.
5. **Mark sure, unsure or guess** honestly. Sure-and-wrong is the most valuable miss on the page — it is a wrong mental model.
6. **Absolutes** (always, never, guarantee) are usually wrong. And recognizing an answer is not the same as owning it.
7. Items with a **heavy black bar** on the left are **transfer items**: the same principle, reversed or solved for a different variable.
8. Stop at the STOP page. Then grade, fill the triage grid, and tally the traps you fell for. The biggest tally is what you drill next.

The strip, filled in, looks like this:

| Target word | The one fact that decides | Answer | [ ] sure [ ] unsure [ ] guess |
|-------------|---------------------------|--------|-------------------------------|
| _____       | _____                     | _____  |                               |

# Items

1.

A teacher at a rural school with a strong robotics club reviews a free online lesson for a standard that asks students to develop a model of the rock cycle showing the flow of energy and cycling of matter. The lesson has students read about rock types and label a rock-cycle diagram. It has a 4.8-star rating from 2,000 users. Is it aligned?

- A.** Yes, because highly rated lessons are aligned.
- B.** Only partly: it covers the topic, but students do not build a model or explain energy and matter flow, so it misses what the standard requires.
- C.** Yes, because it covers rock types and the rock cycle.
- D.** No lesson that includes reading can ever be aligned to a science standard.

| Target word | The one fact that decides | Answer | [ ] sure [ ] unsure [ ] guess |
|-------------|---------------------------|--------|-------------------------------|
| _____       | _____                     | ___    |                               |

2. [TRANSFER — reversed stem / different variable]

A standard asks students to “analyze and interpret data on the distribution of fossils and rocks to provide evidence of past plate motions.” Which task is aligned?

- A.** Students build a baking-soda volcano.
- B.** Students memorize the names of index fossils.
- C.** Students examine maps of matching fossils and rock formations on separate continents and use the data to argue the continents were once joined.
- D.** Students watch a video on Pangaea and answer recall questions.

| Target word | The one fact that decides | Answer | [ ] sure [ ] unsure [ ] guess |
|-------------|---------------------------|--------|-------------------------------|
| _____       | _____                     | ___    |                               |

3.

A science department with a generous textbook budget is choosing between two Earth science programs. Program A’s sales representative says it is “100% standards-aligned” on every page. Program B’s units start from a puzzling phenomenon and have students gather and analyze data to explain it with evidence. Which evaluation approach is sound?

- A.** Use a criteria rubric to look for evidence in the lessons of students using practices to explain phenomena — which favors Program B.
- B.** Choose on price alone, since both say they are aligned.
- C.** Choose the program with the most content pages.
- D.** Choose Program A, since “100% aligned” guarantees alignment.

| Target word | The one fact that decides | Answer | [ ] sure [ ] unsure [ ] guess |
|-------------|---------------------------|--------|-------------------------------|
| _____       | _____                     | ___    |                               |

4.

A teacher in a school that recently repainted its science wing is redesigning a lesson on groundwater. Her class includes students with reading difficulties, students who are far ahead, and several who disengage easily. Which plan best reflects Universal Design for Learning?

- A. Use only hands-on activities, since all students learn best by doing.
- B. Sort students by learning style and teach each group only in its style.
- C. Plan one lesson, then add accommodations only for students with IEPs.
- D. Offer the content through a diagram, a physical model, and text with audio; let students choose a local water problem; and let them show learning by model, writing, or talk.

| Target word | The one fact that decides | Answer | [ ] sure [ ] unsure [ ] guess |
|-------------|---------------------------|--------|-------------------------------|
| _____       | _____                     | ___    |                               |

5. [TRANSFER — reversed stem / different variable]

A colleague says Universal Design for Learning means giving each student a separate lesson matched to his or her learning style. Which response is accurate?

- A. UDL means lowering expectations for struggling students.
- B. UDL builds flexible options into one lesson for all learners from the start; it is not learning-style matching.
- C. UDL applies only to students with disabilities.
- D. That is correct — UDL is learning-style matching.

| Target word | The one fact that decides | Answer | [ ] sure [ ] unsure [ ] guess |
|-------------|---------------------------|--------|-------------------------------|
| _____       | _____                     | ___    |                               |

6.

An Earth science teacher, whose room has a view of the mountains, has several English learners at intermediate proficiency in a unit on weather fronts. The unit task asks students to explain how a cold front produces thunderstorms. Which support keeps the science rigorous while making it accessible?

- A. Wait to teach fronts until the students are fluent in English.
- B. Give no supports, since every student must take the task the same way.
- C. Provide visuals, a word bank, and sentence frames (for example, “When the cold air \_\_\_, the warm air \_\_\_ because \_\_\_”), and accept labeled diagrams.
- D. Replace the explanation with a vocabulary-matching worksheet.

| Target word | The one fact that decides | Answer | [ ] sure [ ] unsure [ ] guess |
|-------------|---------------------------|--------|-------------------------------|
| _____       | _____                     | ___    |                               |

7.

Midway through a unit on the rock cycle, a teacher who coaches the cross-country team gives a quick exit ticket. Most students write that sedimentary rock can form only from igneous rock. Grades close in two weeks. What is the best use of this information?

- A. Record the exit-ticket scores as the unit grade.
- B. Adjust the next lesson to address the misconception — any rock type can weather into sediment — and check again.
- C. Move on as planned; the unit test will catch it.
- D. Reteach the whole unit from the beginning.

| Target word | The one fact that decides | Answer | [ ] sure [ ] unsure [ ] guess |
|-------------|---------------------------|--------|-------------------------------|
| _____       | _____                     | ___    |                               |

8.

A teacher is preparing a slide deck with images from a weather-balloon launch at a science fair. She wants students to explain why temperature rises with height in the stratosphere after falling through the troposphere. The balloon reached 32 km. What is the correct explanation?

- A. Ozone in the stratosphere absorbs ultraviolet light and heats that layer.
- B. Warm air from the surface rises and collects there.
- C. Higher air is closer to the Sun.
- D. Temperature always falls with height throughout the atmosphere.

| Target word | The one fact that decides | Answer | [ ] sure [ ] unsure [ ] guess |
|-------------|---------------------------|--------|-------------------------------|
| _____       | _____                     | ___    |                               |

9. [TRANSFER — reversed stem / different variable]

In the troposphere, how does air temperature generally change with height, and why?

- A. It decreases because the air moves farther from the Sun.
- B. It stays the same, since the air is well mixed.
- C. It decreases, because the troposphere is heated mainly by the ground below.
- D. It increases, because the air is closer to the Sun.

| Target word | The one fact that decides | Answer | [ ] sure [ ] unsure [ ] guess |
|-------------|---------------------------|--------|-------------------------------|
| _____       | _____                     | ___    |                               |

10.

A marine science unit, built around a news story about a lost shipping container of bath toys, asks students to predict how floating objects will drift in the North Pacific. The container held 28,000 toys. Which principle should students use?

- A. The drift is controlled by the tides.
- B. Surface currents are driven mainly by salinity differences.
- C. The gyre turns counterclockwise in the Northern Hemisphere.
- D. Surface currents are driven by prevailing winds and turned by the Coriolis effect into a clockwise gyre in the Northern Hemisphere.

| Target word | The one fact that decides | Answer | [ ] sure [ ] unsure [ ] guess |
|-------------|---------------------------|--------|-------------------------------|
| _____       | _____                     | ___    |                               |

11. [TRANSFER — reversed stem / different variable]

Which way do the large subtropical ocean gyres circulate in the Southern Hemisphere?

- A. Counterclockwise, because salinity differences drive them.
- B. They do not rotate; they flow straight east.
- C. Clockwise.
- D. Counterclockwise.

| Target word | The one fact that decides | Answer | [ ] sure [ ] unsure [ ] guess |
|-------------|---------------------------|--------|-------------------------------|
| _____       | _____                     | ___    |                               |

12.

A teacher at a school that hosts a yearly star party is reviewing a textbook section on the origin of the universe. It says, “The Big Bang was an explosion at a center point in space.” The section has vivid full-page art. Which revision matches current evidence?

- A. The Big Bang was an expansion of space everywhere, not an explosion from a center; evidence includes galaxy redshifts, the cosmic microwave background, and light-element abundances.
- B. Keep it, since redshifts show galaxies flying away from a center in space.
- C. Change it to say the universe is shrinking, since most galaxies are blueshifted.
- D. Change it to say the Big Bang is proven, so no evidence is needed.

| Target word | The one fact that decides | Answer | [ ] sure [ ] unsure [ ] guess |
|-------------|---------------------------|--------|-------------------------------|
| _____       | _____                     | ___    |                               |

# STOP

Answers and dissection follow. Finish every item and every strip first.

# Key and dissection

Struck-through text is the camouflage: facts that do not decide the item.

## 1. Answer B Concept 1

A teacher at a rural school ~~with a strong robotics club~~ reviews a free online lesson for a standard that asks students to develop a model of the rock cycle showing the flow of energy and cycling of matter. The lesson has students read about rock types and label a rock-cycle diagram. ~~It has a 4.8 star rating from 2,000 users.~~ Is it aligned?

**Target word:** alignment judgment **Decides:** The standard asks students to develop a model showing energy and matter.

|          |                   |                                                                 |
|----------|-------------------|-----------------------------------------------------------------|
| A        | Camouflage        | Ratings measure popularity, not alignment.                      |
| <b>B</b> | <b>Correct</b>    | Alignment means students do the practice at the depth required. |
| C        | Dropped element   | Drops the modeling practice and energy/matter explanation.      |
| D        | Absolute language | Reading can support alignment; the gap is the missing practice. |

## 2. Answer C Concept 1 • transfer

A standard asks students to “analyze and interpret data on the distribution of fossils and rocks to provide evidence of past plate motions.” Which task is aligned?

**Target word:** choose the aligned task **Decides:** Students must analyze data as evidence.

|          |                   |                                                              |
|----------|-------------------|--------------------------------------------------------------|
| A        | Adjacent category | A different topic and a demonstration, not data analysis.    |
| B        | Confusion set     | Recall of related vocabulary is not evidence-based argument. |
| <b>C</b> | <b>Correct</b>    | It requires the named practice with the named data.          |
| D        | Dropped element   | Drops the data analysis the standard requires.               |

## 3. Answer A Concept 2

A science department ~~with a generous textbook budget~~ is choosing between two Earth science programs. Program A’s sales representative says it is “100% standards-aligned” on every page. Program B’s units start from a puzzling phenomenon and have students gather and analyze data to explain it with evidence. Which evaluation approach is sound?

**Target word:** how to evaluate **Decides:** Evidence in the lessons vs. a label.

|          |                   |                                                              |
|----------|-------------------|--------------------------------------------------------------|
| <b>A</b> | <b>Correct</b>    | Claims are not evidence; the student work in the lessons is. |
| B        | Dropped element   | Drops the evidence review entirely.                          |
| C        | Adjacent category | Coverage volume is not quality or alignment.                 |
| D        | Absolute language | A marketing label guarantees nothing.                        |

#### 4. Answer D Concept 3

A teacher ~~in a school that recently repainted its science wing~~ is redesigning a lesson on groundwater. Her class includes students with reading difficulties, students who are far ahead, and several who disengage easily. Which plan best reflects Universal Design for Learning?

**Target word:** UDL plan **Decides:** Flexible options built in for all.

|          |                    |                                                                                           |
|----------|--------------------|-------------------------------------------------------------------------------------------|
| A        | Absolute language  | No single mode is best for all.                                                           |
| B        | Confusion set      | Learning-style matching is not UDL and is not supported by research.                      |
| C        | Inverted operation | UDL is proactive design for everyone, not after-the-fact retrofits.                       |
| <b>D</b> | <b>Correct</b>     | Multiple means of representation, engagement, and expression are built in from the start. |

#### 5. Answer B Concept 3 • transfer

A colleague says Universal Design for Learning means giving each student a separate lesson matched to his or her learning style. Which response is accurate?

**Target word:** what UDL is **Decides:** Flexibility in one design vs. style matching.

|          |                   |                                                   |
|----------|-------------------|---------------------------------------------------|
| A        | Adjacent category | UDL keeps goals firm and makes the path flexible. |
| <b>B</b> | <b>Correct</b>    | UDL removes barriers through design, not labels.  |
| C        | Absolute language | UDL is designed for every learner.                |
| D        | Confusion set     | Confuses UDL with the learning-styles idea.       |

#### 6. Answer C Concept 4

An Earth science teacher, ~~whose room has a view of the mountains~~, has several English learners at intermediate proficiency in a unit on weather fronts. The unit task asks students to explain how a cold front produces thunderstorms. Which support keeps the science rigorous while making it accessible?

**Target word:** appropriate language support **Decides:** Keep the causal explanation; support the language.

|          |                   |                                                                     |
|----------|-------------------|---------------------------------------------------------------------|
| A        | Adjacent category | Delay denies access to content; language and content grow together. |
| B        | Absolute language | Equitable access often requires supports.                           |
| <b>C</b> | <b>Correct</b>    | Language supports lower the language barrier, not the thinking.     |
| D        | Dropped element   | Drops the causal reasoning the task is meant to assess.             |

#### 7. Answer B Concept 5

Midway through a unit on the rock cycle, a teacher ~~who coaches the cross-country team~~ gives a quick exit ticket. Most students write that sedimentary rock can form only from igneous rock. ~~Grades close in two weeks.~~ What is the best use of this information?

**Target word:** use of formative data **Decides:** Mid-unit check reveals a misconception.

|          |                   |                                                                          |
|----------|-------------------|--------------------------------------------------------------------------|
| A        | Confusion set     | Treats a formative check as a summative score.                           |
| <b>B</b> | <b>Correct</b>    | Formative data is for adjusting instruction.                             |
| C        | Dropped element   | Drops the step of acting on the evidence.                                |
| D        | Adjacent category | Reteaching is right in spirit, but a targeted fix fits the specific gap. |

## 8. Answer A Concept 6

A teacher is preparing a slide deck with images from a weather-balloon launch ~~at a science fair~~. She wants students to explain why temperature rises with height in the stratosphere after falling through the troposphere. ~~The balloon reached 32 km.~~ What is the correct explanation?

**Target word:** why the stratosphere warms **Decides:** Ozone absorbs UV.

|   |                   |                                                                        |
|---|-------------------|------------------------------------------------------------------------|
| A | <b>Correct</b>    | The heat source is within the layer, near its top.                     |
| B | Adjacent category | Rising air cools as it expands; it does not heat the stratosphere.     |
| C | Confusion set     | The extra distance is negligible compared with the Earth–Sun distance. |
| D | Absolute language | It reverses in the stratosphere and thermosphere.                      |

## 9. Answer C Concept 6 • transfer

In the troposphere, how does air temperature generally change with height, and why?

**Target word:** troposphere temperature trend **Decides:** Heated from the ground up.

|   |                    |                                                                               |
|---|--------------------|-------------------------------------------------------------------------------|
| A | Confusion set      | Right trend, wrong reason; the ground is the heat source.                     |
| B | Dropped element    | Drops the ground-heating effect.                                              |
| C | <b>Correct</b>     | Farther from the heat source and expanding air both mean cooler temperatures. |
| D | Inverted operation | Temperature falls with height in the troposphere.                             |

## 10. Answer D Concept 7

A marine science unit, built around a news story about a lost shipping container of bath toys, asks students to predict how floating objects will drift in the North Pacific. ~~The container held 28,000 toys.~~ Which principle should students use?

**Target word:** what drives surface drift **Decides:** Surface currents and gyres.

|   |                    |                                                                       |
|---|--------------------|-----------------------------------------------------------------------|
| A | Adjacent category  | Tides move water back and forth; they do not create basin-wide gyres. |
| B | Confusion set      | Salinity and temperature drive deep (thermohaline) circulation.       |
| C | Inverted operation | Northern Hemisphere gyres turn clockwise.                             |
| D | <b>Correct</b>     | Wind plus Coriolis shapes the gyre that carries floating objects.     |

## 11. Answer D Concept 7 • transfer

Which way do the large subtropical ocean gyres circulate in the Southern Hemisphere?

**Target word:** gyre direction by hemisphere **Decides:** Coriolis deflects left in the Southern Hemisphere.

|   |                    |                                                                |
|---|--------------------|----------------------------------------------------------------|
| A | Confusion set      | Right direction, wrong cause; winds and Coriolis drive them.   |
| B | Adjacent category  | Continents and Coriolis turn the flow into closed loops.       |
| C | Inverted operation | That is the Northern Hemisphere direction.                     |
| D | <b>Correct</b>     | Leftward deflection turns the southern gyres counterclockwise. |

## 12. Answer A Concept 8

A teacher ~~at a school that hosts a yearly star party~~ is reviewing a textbook section on the origin of the universe. It says, "The Big Bang was an explosion at a center point in space." ~~The section has vivid full-page art.~~ Which revision matches current evidence?

**Target word:** accurate revision **Decides:** Expansion of space everywhere.

|   |                    |                                                                               |
|---|--------------------|-------------------------------------------------------------------------------|
| A | Correct            | All three lines of evidence point to expansion from a hot, dense state.       |
| B | Confusion set      | Redshift increasing with distance shows expansion everywhere, with no center. |
| C | Inverted operation | Most galaxies are redshifted; the universe is expanding.                      |
| D | Absolute language  | Science rests on evidence, and the text should present it.                    |

## Triage grid

Write each item number in the box it belongs to.

|                       | Right                                                 | Wrong                                                                                 |
|-----------------------|-------------------------------------------------------|---------------------------------------------------------------------------------------|
| <b>Sure</b>           | Owled.                                                | <b>Concept inversion — top priority.</b> A wrong mental model; it survives rewording. |
| <b>Unsure / guess</b> | Recognition, not mastery. Re-drill with the transfer. | Not learned yet.                                                                      |

## Trap tally

For every miss, find the letter you chose in the key and tally its trap. The biggest number is the pattern to drill.

| Trap               | What it does                                                 | In this drill | Your misses |
|--------------------|--------------------------------------------------------------|---------------|-------------|
| Dropped element    | Leaves out a required term or step.                          | 6             |             |
| Wrong stem number  | A figure copied from the prompt that is not the answer.      | 0             |             |
| Adjacent category  | The right idea in the wrong bucket.                          | 8             |             |
| Inverted operation | Direction flipped: add/subtract, cause/effect, higher/lower. | 5             |             |
| Absolute language  | Always, never, guarantee — where judgment applies.           | 7             |             |
| Confusion set      | Every option is real; the test is exact role-matching.       | 9             |             |
| Camouflage         | Answers a stem fact that was never the question.             | 1             |             |

# Anchor rows — the concept list

One row per concept: the rule to lock in, the trap that catches people, and the tell in the question that signals it.

| # | Concept                                                   | Rule                                                                                                                                                                                                           | Trap                                                                                      | Tell                                                                                          |
|---|-----------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------|
| 1 | <b>Aligning a resource to a standard</b>                  | A resource is aligned only if students do what the standard asks — the same practice and depth — not just study the same topic.                                                                                | Calling a lesson aligned because it covers the topic.                                     | A standard with a verb like “develop a model” or “analyze data” next to a lesson description. |
| 2 | <b>Evaluating curriculum with evidence</b>                | Judge materials by evidence in the lessons (students using practices to explain phenomena) with a criteria rubric, not by publisher labels, ratings, or page count.                                            | Trusting “100% aligned” claims or popularity.                                             | A marketing claim next to a description of what students actually do.                         |
| 3 | <b>Universal Design for Learning</b>                      | UDL builds flexible options into one lesson from the start: multiple means of engagement, representation, and action and expression, for all learners.                                                         | Matching lessons to “learning styles,” or adding supports only after the fact.            | A mixed-needs class and a choice among lesson plans.                                          |
| 4 | <b>Supporting English learners without lowering rigor</b> | Give language supports (visuals, word banks, sentence frames, diagrams) while keeping the same science thinking demand.                                                                                        | Replacing explanation with vocabulary matching, or waiting for fluency.                   | English learners facing a reasoning task.                                                     |
| 5 | <b>Formative assessment drives next steps</b>             | Checks during learning are used to adjust instruction — targeted reteaching — before the summative assessment.                                                                                                 | Grading a formative check as a final score, or moving on and hoping the test catches it.  | A mid-unit check that reveals a misconception.                                                |
| 6 | <b>Temperature structure of the atmosphere</b>            | The troposphere is heated from the ground and cools with height; the stratosphere warms with height because ozone absorbs ultraviolet light.                                                                   | “It gets colder the higher you go, everywhere,” or “higher is closer to the Sun.”         | Temperature changing direction with altitude.                                                 |
| 7 | <b>Wind-driven surface currents and gyres</b>             | Surface currents are driven by prevailing winds and turned by the Coriolis effect into gyres that run clockwise in the Northern Hemisphere and counterclockwise in the Southern.                               | Crediting surface currents to salinity or tides, or flipping the direction by hemisphere. | Predicting drift of floating objects in an ocean basin.                                       |
| 8 | <b>Evidence for the Big Bang</b>                          | The Big Bang was an expansion of space everywhere, not an explosion from a center; evidence includes galaxy redshifts increasing with distance, the cosmic microwave background, and light-element abundances. | Picturing an explosion at a point in space.                                               | A text or claim describing the Big Bang’s “center.”                                           |

**Course scope (public description):** MAT Secondary Earth Science Curriculum prepares students to develop and evaluate Earth Science curricula for secondary education by exploring the core principles of Earth's processes, materials, and the relationships between the Earth, Sun, and Moon, as well as the broader context of stars, planets,

and the universe. Learners will delve into the hydrosphere and atmosphere, examining their dynamics and the intricate interactions that drive Earth's systems. The course also emphasizes the evaluation and alignment of Earth Science curriculum resources with secondary standards and individual student needs. Learners will gain the skills to critically assess and adapt instructional materials, ensuring they meet diverse learning styles and needs in secondary Earth Science classrooms. Through practical examples and interactive activities, this course equips learners with the knowledge and tools to effectively design, evaluate, and implement a comprehensive Earth Science curriculum.